Hairy cell leukemia

Indolent B-cell leukemia defined by BRAF V600E · splenomegaly with pancytopenia and monocytopenia · **a single purine-analog course gives most patients remissions lasting years**; former HCL variant (now SBLPN) is a distinct, BRAF wild-type disease

Compiled from primary trial reports and consensus guidelines (Hairy Cell Leukemia Foundation consensus, Grever 2017 and 2026 update; NCCN) verified against PubMed, September 2026.

PRESENTATION

- **Rare: <2% of leukemias**; middle-aged adults, strong male predominance
- **Splenomegaly** (often massive) **without adenopathy**, and **pancytopenia with monocytopenia** — the monocytopenia is the clue on a routine CBC
- **Dry tap:** reticulin fibrosis defeats the aspirate — diagnose on the **core biopsy** (interstitial "fried-egg" infiltrate) + **peripheral blood flow**, which finds the few circulating cells
- **Infections** are the cause of morbidity: bacterial, fungal and **atypical mycobacteria** (monocytopenia + neutropenia)
- **Smear:** small-to-medium cells with oval nuclei and **circumferential hairy cytoplasmic projections**

CLASSIC HCL VS MIMICS

	Classic HCL	SBLPN (ex-HCL-v)
CD11c · CD103	**Bright +** · +	+ · +
CD25	**+**	**–**
CD123	**+ (bright)**	usually –
Annexin A1 (IHC)	**+ — 100% specific** *Falini, Lancet 2004*	–
TRAP	+ (historical stain)	variable
BRAF V600E	**~All** (48/48; >95% in 2026 consensus)	**Wild-type**
Other driver	—	**MAP2K1 in ~50%** *Waterfall 2014*
Blood count	Pancytopenia, **low monocytes**	Leukocytosis, monocytes preserved
Purine analog	**Excellent, durable**	Poorer, less durable

HCL score (CD11c, CD103, CD123, CD25): 3–4 = classic. **WHO-HAEM5 dropped "HCL-variant"** and folded it into **splenic B-cell lymphoma/leukemia with prominent nucleoli (SBLPN)**; **ICC 2022 keeps HCL-v as a provisional entity**.

WHEN TO TREAT — CONSENSUS

- **Any of: Hb <11 g/dL, platelets <100,000/µL, ANC <1,000/µL**
- **Symptomatic splenomegaly**, progressive cytopenias, recurrent infection or constitutional symptoms
- **None of these → observe**; many patients sit for years
- **CR:** Hb >11 without transfusion, platelets >100,000, ANC >1,500, no hairy cells on marrow or blood, spleen regressed. MRD by IHC or flow is prognostic but not yet a treatment target
- **Relapse cutoff is 2 years** (raised from 1): relapse ≥2 yr after CR → retreat with purine analog; <2 yr → switch class

IGHV4-34 — the trap. Classic-looking HCL that is **BRAF wild-type**: all 13 IGHV4-34+ cases were wild-type *Xi, Blood 2012*. Usually unmutated, higher WBC, **poor response to cladribine** and shorter PFS and OS *Arons, Blood 2009*; MAP2K1 mutations cluster here too. **A BRAF-negative "HCL" should prompt a second look**.

Treat the infection first. Cladribine and pentostatin deepen neutropenia for weeks, and post-cladribine fever is common — so **defer the purine analog during active infection**. If counts are too low for the infection to clear, a **BRAF inhibitor (vemurafenib ± rituximab) restores counts without myelosuppression** and can bridge to definitive therapy — off-label.

TREATMENT PATHWAY

Newly diagnosed classic HCL

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Indication to treat?
Hb <11, plt <100k, ANC <1,000, or symptoms
none → **observe**

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Active infection?
control it first — defer the purine analog
bridge: vemurafenib ± rituximab (off-label)

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First line
cladribine ×1 course or **pentostatin + rituximab**: concurrent → MRD-free CR 97%
(delayed R also acceptable; see trial below)

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Restage ~6 months
marrow biopsy + counts → CR / PR / MRD

Relapse

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≥2 yr after CR
repeat purine analog (same or the other) **± rituximab**

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<2 yr, or refractory
vemurafenib + rituximab (CR 87%)
dabrafenib + trametinib (off-label in HCL)
ibrutinib · pentostatin + R · trial

SBLPN (HCL-v) · IGHV4-34 HCL

↓

BRAF wild-type — a different disease
cladribine + **concurrent rituximab** (CR 95%)
BRAF inhibitors have no target
relapse: ibrutinib, trials

Hairy cell leukemia

KEY EVIDENCE

Study	Result	Read it this way
Scripps cladribine *n=358*	Single 7-day infusion: **CR 91%**; 26% relapsed at a median 29 mo; **retreatment CR 62%**	The regimen that made HCL a one-course disease
Else, long-term *n=233, 16 yr f/u*	Pentostatin or cladribine: CR 80%, **median RFS 16 yr**; no difference between drugs	Relapse continues late — 25% of 5-yr CRs relapse by 15 yr
Intergroup *Grever 1995*	**Pentostatin vs interferon-α:** CR **76% vs 11%**. 10-yr OS 81% (Flinn 2000)	Pentostatin is equivalent to cladribine; no excess second cancers
SAKK 32/98 *n=100, randomized*	Cladribine **daily ×5 vs weekly ×5**: response 86% in both	**Weekly is not less toxic** (grade 3–4 neutropenia 90% vs 80%)
NIH, Chihara *n=68, randomized ph2*	Cladribine + **concurrent vs delayed (≥6 mo) rituximab**: CR 100% vs 88% (NS); **MRD-free CR 97% vs 24%**; at 8 yr **94% vs 12%** still MRD-free	Concurrent R buys deep, lasting MRD negativity at the cost of grade 3–4 thrombocytopenia **59% vs 9%**. Survival benefit not shown
MDACC *n=139, single-arm*	Cladribine → rituximab: **CR 97%**, 10-yr EFS **87%**, OS **91%**	Frontline classic HCL relapse only 3.6%
Vemurafenib *Tiacci 2015*	Relapsed: ORR **96–100%**, CR 35–42%; RFS **19 mo** in CR, 6 mo in PR	High response, short remission alone
Vem + R *Tiacci NEJM 2021, n=30*	Median 3 prior lines: **CR 87%** (65% MRD-negative); **PFS 78%** at median 37 mo	The preferred non-purine regimen at relapse
Vem + obinutuzumab *frontline, n=30*	**CR 90%**, 96% of CRs MRD-negative; no relapse at 35 mo	Chemo-free frontline option; single-arm
Dab + tram *ROAR, n=55*	Relapsed: ORR **89%**, CR **66%**; 24-mo PFS 94%	**Not FDA-approved for HCL** — the tumor-agnostic label covers solid tumors only
Ibrutinib *n=37*	Best ORR **54%**; 36-mo PFS 73%	**Primary endpoint not met**; modest
Moxetumomab *n=80*	Anti-CD22 immunotoxin: CR 41%, durable CR 30%; HUS 7.5%, capillary leak 5%	**Withdrawn from the US market in 2023** (commercial decision)
SBLPN: clad + R *n=20*	Concurrent rituximab: **CR 95%**, 5-yr PFS **63%**	TP53-mutant cases fare worse

DOSING

Regimen	Dosing
Cladribine	**0.14 mg/kg/day IV over 2 h, days 1–5** (NIH: 0.15 mg/kg), **or 0.1 mg/kg/day continuous IV ×7 days**, or 0.14 mg/kg weekly ×5 — **one course**
Clad + rituximab	Cladribine 0.15 mg/kg days 1–5 + **rituximab 375 mg/m² weekly ×8**, concurrent from day 1 (or delayed ≥6 mo)
Pentostatin	**4 mg/m² IV every 2 weeks** until best response; hydrate; hold for renal dysfunction
Vemurafenib + R	**Vemurafenib 960 mg PO BID ×8 weeks** + **rituximab 375 mg/m² ×8 doses over 18 weeks**
Dabrafenib + trametinib	Dabrafenib 150 mg BID + trametinib 2 mg daily
Ibrutinib	420 mg daily

MONITORING AND SUPPORTIVE CARE

- **Prolonged CD4 lymphopenia** after purine analogs — median **~40 months** to recover → **PJP and HSV/VZV prophylaxis**
- **Irradiated blood products indefinitely** after a purine analog (transfusion-associated GVHD) *BSH 2020*
- **Neutropenic fever in ~42%** after cladribine. **Routine G-CSF not recommended**: it shortened neutropenia (9 vs 22 days) but did not cut fever or admissions
- **Vemurafenib:** rash, photosensitivity, arthralgia, **cutaneous SCC** / **keratoacanthoma** — dermatology checks. **Dab/tram:** pyrexia
- Inactivated vaccines before therapy; **no live vaccines**

SPLENIC SMALL B-CELL DIFFERENTIAL

Entity	Distinguishing features
Classic HCL	CD25+ CD103+ CD123+ CD11c bright, **annexin A1+**, **BRAF V600E**; monocytopenia
SBLPN (HCL-v)	CD103+ but **CD25–**, annexin A1–, BRAF wild-type, MAP2K1; leukocytosis, prominent nucleoli
Splenic marginal zone	**Polar (villous) projections**, CD103 usually –, CD25 variable, annexin A1–; **NOTCH2** / **KLF2**; may have a small IgM paraprotein
Splenic diffuse red pulp	Villous cells, red-pulp pattern, CD25–, annexin A1–; **CCND3** mutations; BRAF wild-type
CLL · MCL	CD5+ — CD23+ (CLL) or cyclin D1+ (MCL) — HCL and its look-alikes are CD5–

Annexin A1 + BRAF V600E together settle classic HCL; everything else in this list is negative for both.

PEARLS

- **Splenomegaly + monocytopenia + dry tap = HCL** until proven otherwise
- **BRAF V600E is the diagnosis and the target.** Wild-type → think SBLPN or IGHV4-34
- **Cure-like, not cure:** relapse keeps happening late but responds to retreatment
- **Moxetumomab is gone** (2023) — don't list it as an option